Kevin Cornelio Sawatzky

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Education

August 2024 – Present
Ph.D in Nuclear, Plasma & Radiological Engineering, University of Illinois Urbana-Champaign.
Graduate Concentration in Computational Science and Engineering
GPA: 4.0 / 4.0

May 2022 – April 2024
M.ASc. in Nuclear Engineering, Ontario Tech University.
Thesis title: *Coupled Radiation Transport and Mobile Depletion for Health Physics and Environmental Impact Applications.*
GPA: 4.18 / 4.3

September 2017 – April 2022
B.Eng. in Nuclear Engineering, Ontario Tech University.
Minor in Computational Science.
GPA: 3.82 / 4.3

Employment History

August 2024 – Present
Graduate Research Assistant, Nuclear, Plasma & Radiological Engineering, University of Illinois Urbana-Champaign.
- Developing adaptive mesh refinement methods and functional expansion tallies for unstructured mesh Monte Carlo particle transport.
- Developing neutronics capabilities in Cardinal to enhance advanced reactor analysis methods and workflows.

May 2025 – August 2025
Research Aide Technical - PhD, Advanced Fuels and Materials, Argonne National Laboratory.
- Investigated the effects of TRISO particle homogenization on heat transport in TRISO fueled microreactors.
- Developed multi-scale methods to improve predictions of TRISO kernel temperatures during microreactor transients.

May 2024 – July 2024
Research Assistant, Engineering Physics, McMaster University.
- Developed methodologies to analyze surrogate TRISO fuel particles with HR(S)TEM and FIB-SEM instruments at the Canadian Center for Electron Microscopy.
- Co-authored a journal manuscript on high-resolution microscopy methods for TRISO fuel.

May 2022 – April 2024
Graduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
- Developed a discrete ordinates radiation transport solver and finite volume fluid activation/depletion solver in the MOOSE framework.
- Coupled both solvers together for the purposes of analyzing fluid depletion in the health physics and environmental impact code CARIBOU.
- Managed high performance computing resources within the Atkinson lab.

Employment History (continued)

September 2022 – December 2023

Teaching Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Courses taught: *Nuclear Plant Operations, Nuclear Reactor Kinetics, Medical Imaging*

May 2021 – August 2021

Undergraduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Project title: *Literature Review of Small Modular Reactor Instrumentation and Control Strategies*.
- Performed a literature review on Small Modular Reactor instrumentation and control strategies with a focus on integral Pressurized Water Reactors and High Temperature Gas-Cooled Reactors.
- Assisted in preparing a manuscript for journal submission on the results of the literature review.

May 2020 – August 2020

Undergraduate Research Assistant, Department of Energy and Nuclear Engineering, Ontario Tech University.
Project title: *Development of a Radioisotope Atmospheric Dispersion Model in the MOOSE Finite Element Framework*.
- Performed a literature review on methods used for modeling radionuclide dispersion.
- Developed a MOOSE mini-application to perform atmospheric dispersion calculations, simulating the first 24 hours of the Chernobyl plume.

Research Interests and Specialties

Radiation Transport Methods

Neutron transport, charged particle transport, discrete ordinates methods, Monte Carlo methods and variance reduction, code verification and validation.

Multi-Physics Methods

Coupled neutronics, radionuclide depletion and thermal-hydraulics; multi-physics verification and validation.

Advanced Reactor Simulation

Modeling pebble bed, molten salt, liquid metal cooled, and prismatic gas-cooled reactors.

Publications

Journal Articles

1. **K. C. Sawatzky**, A. J. Novak, and K. D. Atkinson, "A Self-Adjoint Scalar Flux Equation for Calculating Point Source Uncollided Flux Moments Without Ray Effects," *Nuclear Science and Engineering*, 2025. DOI: https://doi.org/10.1080/00295639.2025.2592177.

Technical Reports

1. Y. Miao, S. Lee, **K. Sawatzky**, *et al.*, "Multiphysics Modeling of Microreactors with NEAMS codes, and Validation Based on KRUSTY Reactivity Insertion," Argonne National Laboratory, ANL/NEAMS-25/2, 2025.
2. B. Wirth, A. Novak, E. Merzari, W. Brinkley, and **K. Sawatzky**, "Developing a Science-based Predictive Simulation Capability for the Design and Analysis of Advanced Nuclear Fuel Concepts Operating in a Virtual Reactor Environment," Sawtooth Simulation, DE-SC0025110, 2025.

Conference Proceedings

1. E. W. Ahammed, **K. Sawatzky**, L. Gross, *et al.*, "Development of Mesh Tally Amalgamation Algorithm for Coupled High Fidelity Multiphysics Simulation," in *Proceedings of PHYSOR 2026 (submitted)*, Turin, Italy: Politecnico di Torino, 2026.
2. M. A. Mendes, M. Eltawila, **K. C. Sawatzky**, A. J. Novak, and R. C. Martineau, "Validation of Cardinal Coupled Neutronics and Thermomechanics Using the KRUSTY Experiments," in *Proceedings of NETS 2026 (accepted)*, Dayton, Ohio, USA: American Nuclear Society, 2026.
3. C. H. H. Moore, **K. C. Sawatzky**, A. J. Novak, and K. D. Atkinson, "Validation of a Two-Step Neutronics Workflow using Gnat and OpenMC with the ZED-2 Research Reactor," in *Proceedings of PHYSOR 2026 (submitted)*, Turin, Italy: Politecnico di Torino, 2026.
4. **K. C. Sawatzky**, E. W. Ahammed, P. P. Wilson, P. C. Shriwise, P. K. Romano, and A. J. Novak, "Mesh Tally Adaptive Mesh Refinement Strategies for Spatial Gradients Driven by Material Heterogeneities," in *Proceedings of PHYSOR 2026 (submitted)*, Turin, Italy: Politecnico di Torino, 2026.
5. **K. C. Sawatzky**, A. J. Novak, and Y. Miao, "Application of a Multiscale Temperature Treatment to Multiphysics Calculations in a TRISO Fueled Microreactor," in *Proceedings of PHYSOR 2026 (submitted)*, Turin, Italy: Politecnico di Torino, 2026.
6. **K. C. Sawatzky**, E. W. Ahammed, P. P. Wilson, P. C. Shriwise, P. K. Romano, and A. J. Novak, "Adaptive Mesh Refinement Applied to Unstructured Mesh Tallies in Cardinal-OpenMC," in *Proceedings of ANS-2025*, Chicago, Illinois, U.S.A: American Nuclear Society, 2025.
7. **K. C. Sawatzky** and A. J. Novak, "Heuristic-Based Adaptive Mesh Refinement Applied to Ray Effect Mitigation Methods," in *Proceedings of M&C-2025*, Denver, Colorado, U.S.A: American Nuclear Society, 2025.
8. **K. Sawatzky** and K. D. Atkinson, "Verification of a CARIBOU-OpenMC Workflow for the Analysis of HTGR-Like Systems Using the Proposed Ontario Tech Subcritical Assembly," in *Proceedings of PHYSOR-2024*, San Francisco, California, U.S.A: American Nuclear Society, 2024.
9. **K. Sawatzky** and K. D. Atkinson, "Development of a Near-Field Neutron Activation and Dispersion Solver for the Environmental Impact Code CARIBOU," in *Proceedings of M&C-2023*, Niagara Falls, Ontario, Canada: Canadian Nuclear Society, 2023.

Skills

Programming Languages	C, C++, Python, LaTeX, Java, MATLAB, FORTRAN
Frameworks	MOOSE, FEniCS
Tools	OpenMC, Cardinal, Siemens NX, coreform Cubit, DRAGON3/5, nekRS
Miscellaneous	ParaView, git, MS Office suite